Khalil Peguero

954-412-9214 | kpeguero@fiu.edu | linkedin.com/in/khalilpeguero | github.com/Kpeguero16

Education

Florida International University — Miami, FL

Bachelor of Science in Computer Science — *Expected Summer 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Algorithm Techniques, Systems Programming, Programming Languages & Compiler Theory, Data Mining & Machine Learning, Probability & Statistics

Experience

Swarms — Remote

Agent Engineer Intern — *October 2026 – Present*

- Merged **23 PRs** (**17** code, **6** docs) for an open-source **Python** multi-agent framework (**7K+** GitHub stars), **6** over competing PRs, adding email channels and a **GraphWorkflow** error policy and removing **430+** lines of dead code.
- Built **SMTP/IMAP** and **AgentMail** (**REST**, **httpx**) channels that keep agents and people in one email thread; live API tests exposed truncated agent posts, fixed with a length header (**18/18** bodies read back exactly).
- Fixed **CronJob.stop()** failing with "cannot join current thread" when a job stopped itself, which left its task scheduled, and a restart race; added **6** sleep-free regression tests hardened against **16** deliberately faulty versions of the method.

Sigma Lambda Beta Education Foundation, Inc. — Remote

Software Engineer — *May 2025 – Present*

- Maintain an **ASP.NET/C#** and **MySQL** donation portal (**27K+** donations) with **Authorize.NET** payments.
- Consolidated payment types via an idempotent, self-validating **SQL** migration with rollback, tested against **24** scenarios and a production snapshot; fixed a bug where editing a contribution silently overwrote its payment type.
- Refactored the District model into Region across the **MySQL** schema, entity layer, services and reports (**30+** files), fixing **2** pre-existing crashes found in QA and halving the region report's database round-trips.
- Authored **C#** unit tests for the contribution validation service (**98.7%** line coverage) and integration tests for bug fixes, run by **CI** on every commit; peer-reviewed **12** teammate PRs spanning **PCI** remediation and **OAuth** flows.

BNY — Remote

Sophomore Summit – Tech Lead — *May 2025 – July 2025*

- Led a **5-member team** in a mock-IPO challenge; presented a **Python** share-allocation algorithm to senior engineers.

Domus Care Solutions — Weston, FL

Software Engineer Intern — *May 2024 – July 2024*

- Built a **Java Swing** desktop report builder that formats sensitive data for **hundreds of patients** into internal reports.
- Automated **Google Sheets** dashboards for patient activity, billing and claims, cutting manual sorting by **8+ hours/week**.

Projects

MomentumX — Tokenized Stock Trading Platform | *Python, FastAPI, SQLite, Solana, pytest* — September 2026

- Won **MLH Best Use of Solana** at **ShellHacks 2026** (294 projects) as backend lead of a 4-person team.
- Built the **Solana** transaction layer in **FastAPI**: verifies each wallet-signed trade against its quote, then co-signs and confirms it; made submission **idempotent** so dropped connections never lose or double a trade.
- Designed an average-cost P/L ledger that serves portfolio stats in ∼**11 ms**; fixed intermittent **HTTP 500s** under polling via per-thread SQLite connections and **WAL**, cutting load-test errors from **50/900 to 0**.

QuantSim — Paper Trading & Backtesting Platform | *Go, PostgreSQL, Redis, Docker, React* — February 2026 – Present

- Architected a **Go** microservices platform (auth, gateway, market data, trading, backtesting, analytics) with **600+ tests**.
- Load-tested the full order path (gateway, JWT auth, cached price, Postgres transaction) at ∼**2,000 orders/s** with **64 ms** p99 at **100** concurrent clients; reconciled **500K+** fills against the ledger with **0** balance or position mismatches.
- Row locks (**SELECT FOR UPDATE**) stop double-spends: **500** concurrent orders, balance for **10**, exactly **10** filled.
- Fixed gateway port exhaustion (**60%** 502s at **100** clients) with connection pooling: reads **1.3K to 25.7K req/s**, **0** errors.
- Built MA/RSI/MACD backtests (look-ahead-safe, Sharpe/drawdown) at ∼**5M bars/s** (**2.5M** bars, **100** symbols, **0.5 s**).
- Added **Claude API** summaries with a hallucination guard: Go injects every figure and rejects drafts containing digits.

Leadership & Involvement

- **President** (2024–2025) and **VP of Finance** (2024–Present), FIU chapter of Sigma Lambda Beta International Fraternity; coordinated a joint emergency fundraiser with Sigma Lambda Gamma for Venezuela earthquake relief.
- **Member**, INIT and SHPE

Technical Skills

Languages: Python, Go, Java, C#, C, SQL, TypeScript, JavaScript, HTML/CSS
Frameworks & Tools: FastAPI, ASP.NET/.NET, React, Node.js, pytest, LiteLLM, Git, Claude Code, Codex
Data & Infrastructure: PostgreSQL, MySQL, SQLite, Redis, Docker, Linux, CI/CD, Solana
Engineering Practices: multi-agent systems, REST APIs, microservices, unit/integration/mutation testing, JWT/OAuth2